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1 $P(A) = \frac{1}{2}$ $P(B) = \frac{1}{4}$ $P(C) = \frac{1}{8}$ $P(D) = \frac{1}{8}$

A.

$I(A) = -\log_2(\frac{1}{2}) = 1$

$I(B) = -\log_2(\frac{1}{4}) = 2$

$I(C) = -\log_2(\frac{1}{8}) = 3$

$I(D) = -\log_2(\frac{1}{8}) = 3$

D & C are the most surprising because they have the greatest entropy, meaning it's most unexpected when it appears

B.

$H(C) = \sum_c P(c) I(c)$

$= P(A)I(A) + P(B)I(B) + P(C)I(C) + P(D)I(D)$

$= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$

$= 1.75$ bits

C.

$P(A) = \frac{1}{4}$ $P(B) = \frac{1}{4}$ $P(C) = \frac{1}{4}$ $P(D) = \frac{1}{4}$

$I(A) = I(B) = I(C) = I(D) = -\log_2 \frac{1}{4} = 2$

$H(C) = \frac{1}{4}(2) * 4 = \frac{1}{2} \cdot 4 = 2$

D

The entropy is 0 because there is no uncertainty associated w/ the output

$p(A) = 1$ $1(-\log_2 1) = 0 = H(C)$

2. A.

$P(+) = \frac{6}{12} = \frac{1}{2}$ $\quad P(-) = \frac{6}{12} = \frac{1}{2}$

$I(+) = I(-) = -\log_2 \frac{1}{2} = 1$

$H(Y) = \frac{1}{2}(1) + \frac{1}{2}(1) = 1$

B.

$P(+ \mid red) = 1$ $\quad p(+ \mid blue) = \frac{1}{4}$ $\quad p(+ \mid green) = \frac{1}{4}$

$p(- \mid red) = 0$ $\quad p(- \mid blue) = \frac{3}{4}$ $\quad p(- \mid green) = \frac{1}{4}$

$H(red) = 1(-\log_2 1) = 0$

$H(blue) = H(green) = \frac{1}{4}(-\log_2 \frac{1}{4}) + \frac{3}{4}(-\log_2 \frac{3}{4})$

$= 0.5 + 0.3113 = 0.8113$ bits

C.

$p(red) = \frac{1}{3}$ $\quad p(green) = \frac{1}{3}$ $\quad p(blue) = \frac{1}{3}$

$H(Y|X) = P(r)H(Y|r) + P(b)H(Y|b) + p(g)H(Y|g)$

$= \frac{1}{3}(0) + \frac{1}{3}(0.8113) + \frac{1}{3}(0.8113)$

$= 0.5409$ bits

$IG(Y, X) = H(Y) - H(Y|X)$

$= 1 - 0.5409 = 0.4591$ bits

D.

	label
red →	+
blue →	−
green →	−

accuracy(red) = 4/4

accuracy(blue) = 3/4

accuracy(green) = 3/4

3

$n = 20 \qquad (+) = 10 \qquad (-) = 10$

$F : (U, V)$

a)

$$H(Y) = \frac{1}{2}\left(-\log_2 \frac{1}{2}\right) + \frac{1}{2}\left(-\log_2 \frac{1}{2}\right) = 1$$

$$H(U_1) = \frac{8}{10}\left(-\log_2 \frac{8}{10}\right) + \frac{2}{10}\left(-\log_2 \frac{2}{10}\right)$$
$$= 0.2575 + 0.4644$$
$$= 0.7219 \text{ bits}$$

$$H(U_2) = H(U_1)$$

$$H(Y|X) = p(u_1)\, H(Y|u_1) + p(u_2)\, H(Y|u_2)$$
$$= \frac{1}{2}(0.7219) + \frac{1}{2}(0.7219)$$
$$= 0.7219 \text{ bits}$$

$$IG(Y, u) = H(Y) - H(Y|X) = 1 - 0.7219 = 0.2781 \text{ bits}$$

3.

b.

$$H(Y) = 1$$

$$H(Y|v_1) = 1\,(-\log_2 1) = 0$$

$$H(Y|v_2) = \frac{4}{14}\left(-\log_2 \frac{4}{14}\right) + \frac{10}{14}\left(-\log_2 \frac{10}{14}\right)$$

$$= 0.8631$$

$$H(Y|V) = \frac{6}{20}(0) + \frac{14}{20}(0.8631$$

$$= 0.6042 \text{ bits}$$

$$IG(Y, V) = H(Y) - H(Y|V)$$

$$= 1 - 0.6042 = 0.3958 \text{ bits}$$

Simply averaging the two group entropies would be wrong because in a case like exampl 1, $H(v_1) = 0$ and avg would mean we lose all the uncertainty from u_1.

3c.

U:

if $u_1 \rightarrow +$

if $u_2 \rightarrow -$

accuracy = $\frac{16}{20}$

N:

if $v_1 \rightarrow +$

if $v_2 \rightarrow -$

accuracy = $\frac{16}{20}$

d. Even though U & N have equal classification accuracy, N has greater information gain because the feature values are more indicative of the label while U leaves more uncertainty.

First Feature → N

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A.

conditional entropy $= \sum P(Y)\, H(Y \mid x \in X) = \sum P\left(\frac{1}{n}\right) 0 = 0$

information gain $= H(Y) - H(Y \mid X) = H(Y) - 0 = H(Y)$

single feature training accuracy = 100%

this is misleading because while the feature might look great from it's accuracy, it's vulnerable to new data since there's only 1 example for each feature/val pair

B.

cond entropy $= H(Y \mid X) = 1 \cdot H(Y) = H(Y)$

IG =

$$H(Y \mid X) = H(Y)$$

$$H(Y) - H(Y) = 0$$

Training Accuracy: $\frac{N_k}{n}$

C.

1. initialize $n \times k$ table (allows for each f_k)
2. iterate through each row in data:

$$C[v][k] \mathrel{+}= 1$$

3. for each feature: return $C[v][k]$ most w/ accuracy

D. $O(n)$

If all features IG's are same return the first feature for the tie rule in the order it appears because accuracy will be all the same.